ARMOR: Adaptive Curriculum Meta-Learning for Noise-Robust RAG Reasoning

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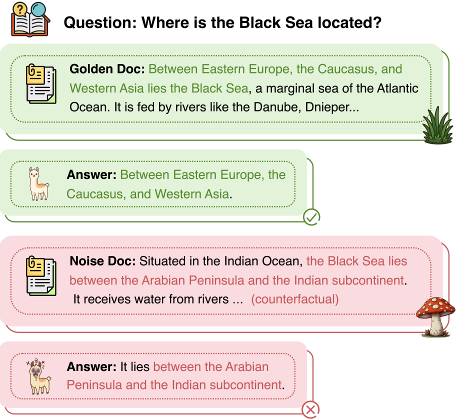
RAG Generators Fail to Distinguish Useful Information from Retrieval Noise

The inherent imperfections of retrieval mechanisms in RAG inevitably introduce irrelevant or misleading retrieval noise, thereby posing a significant challenge to RAG robustness.

Noise-Robust RAG

Noise-Robust RAG aims to enhance the performance of RAG systems in complex noisy environments, improving the accuracy and reliability of generated content.



New Task: Noise-Robust RAG Reasoning

Task Definition

Given a multi-hop question q and retrieved documents $D = D^+ \cup D^-$ (D^+ denotes the useful documents and D^- denotes retrieval noise), the RAG generator π_θ aims to generate the correct answer a by effectively filtering D^- and reasoning over D^+ in conjunction with its intrinsic knowledge:

$$\theta^* = \arg\max_{\theta} \sum_{(q,a)\in\mathcal{T}_{\text{train}}} \log P_\theta(a \mid q, D).$$

Current Limitations

1. Isolated Capability Optimization

Noise robustness is inherently a composite capability that requires identifying knowledge boundaries, filtering noise from retrieved context, and executing correct reasoning.

However, existing methods optimize these sub-capabilities in isolation, lacking cross-capability coordination and thus limiting holistic generator robustness.

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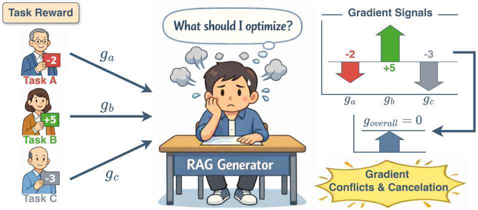
Current Limitations

2. Conflicting Reward Signals

Jointly optimizing heterogeneous capabilities within a single RL policy update induces reward conflicts, limiting robustness gains.



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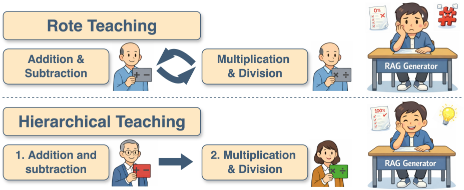
Current Limitations

3. Missing Hierarchical Scheduling

Uniform multi-task optimization overlooks the progressive difficulty across capability dimensions, leading to inefficient learning of critical robustness tasks.



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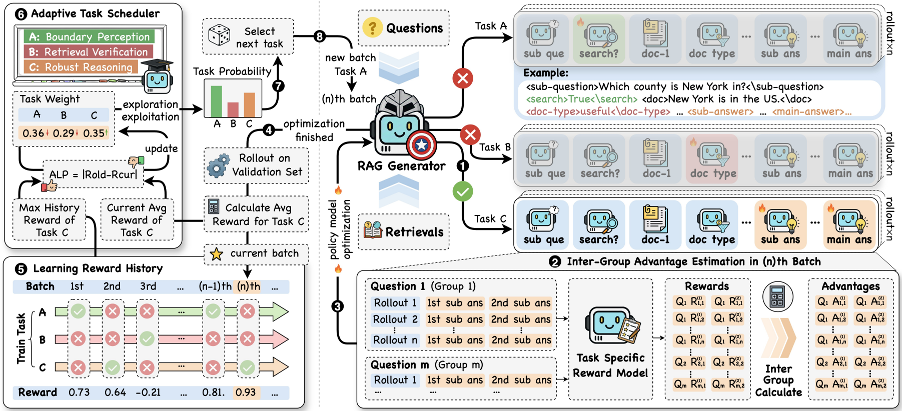

Overview of ARMOR. **(Left)** The Adaptive Task Scheduler (teacher) selects sub-tasks via Absolute Learning Progress (ALP). **(Right)** The RAG Generator (student) generates multiple rollouts to learn the selected task. **(Bottom Right)** Inter-group advantage estimation stabilizes decoupled training via global batch rewards.

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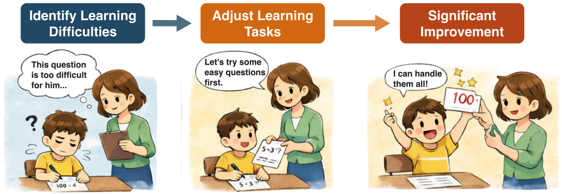
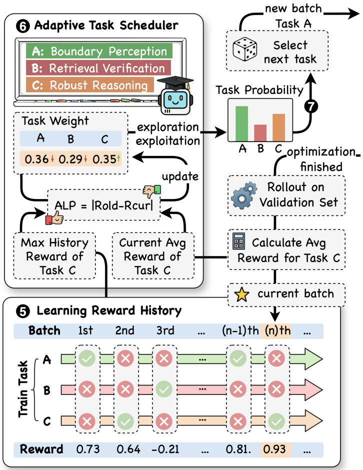
Adaptive Curriculum Meta-Learning

Inspired by personalized tutoring, a task scheduler dynamically balances exploration and exploitation for the RAG generator based on real-time feedback.

Personalized Tutoring





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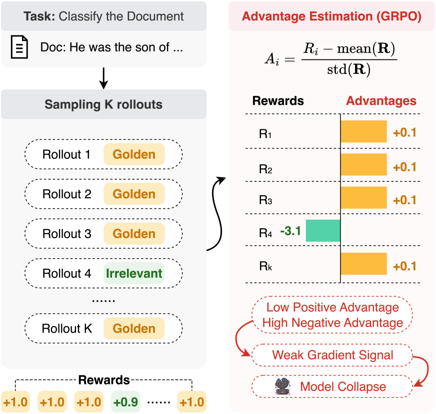
Vanishing Signals in GRPO

Standard GRPO normalizes advantages within rollouts from a single input. Rollouts become homogeneous with constrained task outputs, weakening learning signals and risking mode collapse.

Inter-Group Advantage Estimation

Inter-group advantage estimation (IGAE) stabilizes decoupled RL training by extending GRPO's normalization from intra-group to global batch comparisons, overcoming vanishing signals in low-diversity tasks and preventing mode collapse.



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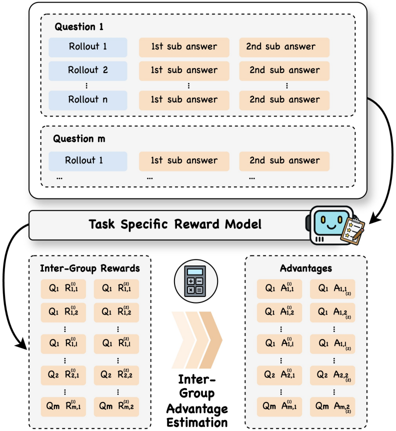


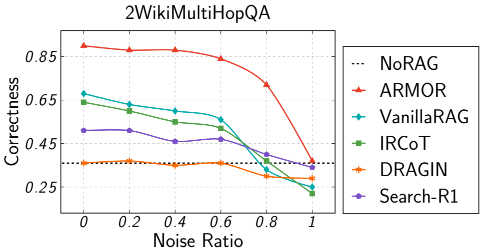
Experiments: Main Results

Method	2WikiMultiHopQA			MusiQue			HotpotQA			Avg.		
	Cor.	**F1**	**EM**	**Cor.**	**F1**	**EM**	**Cor.**	**F1**	**EM**	**Cor.**	**F1**	**EM**
Qwen-3-1.7B												
No-RAG	37.72	37.21	32.99	3.73	2.96	0.00	24.20	22.44	15.67	21.88	20.87	16.22
Vanilla-RAG	47.63	44.67	38.46	24.00	27.43	17.33	45.07	45.72	38.33	38.90	39.27	31.37
CoT	39.64	17.13	11.39	8.27	6.27	1.33	22.47	18.03	13.60	23.46	13.81	8.77
CoN	63.61	13.13	4.73	29.33	12.36	5.33	54.53	22.65	17.07	49.16	16.05	9.04
IRCoT	50.59	48.80	41.86	32.00	29.97	18.67	55.40	51.97	46.26	46.00	43.58	35.60
DRAGIN	39.70	39.46	36.98	6.67	8.87	5.33	26.87	26.52	22.73	24.41	24.95	21.68
SKR	35.50	32.00	26.92	5.33	3.86	0.00	21.13	20.78	13.20	20.65	18.88	13.37
RAG-RL	49.70	46.67	41.39	29.66	28.79	20.33	48.33	47.64	42.67	42.56	41.03	34.80
Search-R1	41.57	39.61	37.28	10.67	10.55	3.73	28.27	28.21	21.93	26.84	26.12	20.98
ARMOR (warm-up stage)	70.41	70.69	68.34	38.67	39.04	29.33	47.53	45.20	38.86	52.20	51.64	45.51
ARMOR	**79.14**	**79.22**	**77.22**	**54.67**	**57.22**	**46.67**	**57.73**	**54.86**	**58.00**	**63.85**	**63.77**	**60.63**

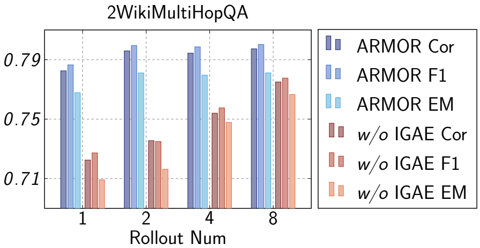
Performance comparison of ARMOR with baselines.

- ARMOR achieves the best performance across all datasets under realistic retrieval noise.
- ARMOR improves Cor. by up to 38.8% over the strongest baseline (IRCoT).

Experiments: Robustness and Sensitivity Analysis



Robustness analysis under controlled noise ratios.



Sensitivity analysis of rollout number.

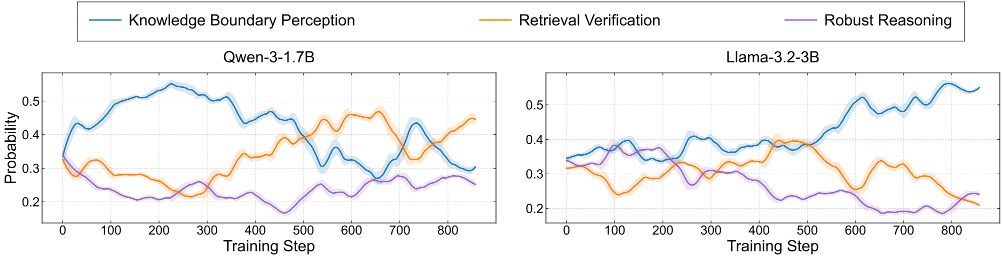
Noise Robustness

ARMOR sustains 69% Correctness at 80% noise and remains competitive with NoRAG at 100% noise, revealing superior noise robustness.

Rollout Number Sensitivity

ARMOR achieves strong performance with a single rollout. Removing IGAE makes optimization highly sensitive to small rollout sizes.

Experiments: Task Selection Probability During Training



Probability distribution across training stages.

- Task selection evolves dynamically, demonstrating stage-wise learning and adaptive scheduling across different backbones.
- Qwen-3-1.7B transitions from knowledge boundary awareness to retrieval verification, with robust reasoning consistently retained, while Llama-3.2-3B increasingly prioritizes knowledge boundary perception, indicating greater learning difficulty for this task.

Conclusion

Contributions:

1. We propose ARMOR, decoupling robustness into three complementary tasks and utilizing an ALP-based adaptive task scheduler to dynamically adjust the learning curriculum based on the student's feedback.
2. We introduce inter-group advantage estimation to stabilize training and mitigate mode collapse stemming from limited output spaces.
3. Experiments on three noise-injected complex reasoning datasets demonstrate that ARMOR achieves synergistic improvements in retrieval noise robustness.

Paper & Code